

browser-use on Amazon Bedrock

Testing report · `persona-browser-use` agent · game-starclash live Playground trial

BEDROCK AUTH: FIXED

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Goal

Make the Dockerized `persona-browser-use` agent authenticate with **Amazon Bedrock** so a live Playground trial can run on a host that only has a Bedrock bearer token (`AWS_BEARER_TOKEN_BEDROCK`) — no direct `ANTHROPIC_API_KEY` and no AWS IAM keys. This is a **testing-only** change to the shared browser-use agent adapter; it is **not** part of the game-starclash task PR.

✓ **Result.** Before the change, the agent was rejected before it started ("Set `ANTHROPIC_API_KEY` ... for browser-use"), then failed every model call with "Could not resolve authentication method". **After the change the agent authenticates through Bedrock and makes real model round-trips** — the log now reads `provider=anthropic_bedrock` and `model=us.anthropic.claude-haiku-4-5-20251001-v1:0`, and the model returns reasoning + a final action. The Bedrock auth blocker is resolved.

Why it was blocked (root cause)

There were **two** layers, both broken for a Bedrock-only host:

Layer	Problem
<code>browser_use.py</code> (adapter)	Hard-required one of <code>ANTHROPIC_API_KEY</code> / <code>OPENAI_API_KEY</code> / <code>LLM_API_KEY</code> / <code>DASHSCOPE_API_KEY</code> and never forwarded any AWS/Bedrock env into the container → rejected up front.
<code>browser_use_runner.py</code> (in-container)	<code>_create_llm()</code> only ever built <code>ChatAnthropic</code> (direct <code>api.anthropic.com</code>). No branch for <code>ChatAnthropicBedrock</code> → Bedrock was impossible even if creds were present.
anthropic SDK pin	<code>browser-use</code> 0.13.6 pins <code>anthropic==0.76.0</code> , whose Bedrock client only signs with SigV4 (IAM access keys) and cannot use a bearer token. A Bedrock-key-only host raises "could not resolve credentials from session".

The tell-tale progression (same host, three trials)

```
# Trial 0 (pre-change) — rejected before launch
ValueError: Set ANTHROPIC_API_KEY, OPENAI_API_KEY, DASHSCOPE_API_KEY, or LLM_API_KEY for browser-use

# Trial 1 (adapter + runner branch, SDK still 0.76.0) — Bedrock selected, SigV4 fails
INFO [Agent] Starting ... provider=anthropic_bedrock and model=us.anthropic.claude-haiku-4-5-...
WARNING [Agent] LLM error (ModelProviderError: could not resolve credentials from session) # x6

# Trial 2 (also upgrade anthropic>=0.115 in the venv) — bearer token accepted, real round-trips
INFO [Agent] Starting ... provider=anthropic_bedrock and model=us.anthropic.claude-haiku-4-5-...
INFO [Agent] done: text: "The task involves playing a Rock-Paper-Scissors game ..." success: False # model answered
```

Proof the bearer token is the deciding factor: the project venv ships `anthropic 0.115.0` and a direct `AnthropicBedrock(...).messages.create(...)` on this host returns PONG live. The container venv shipped `0.76.0`, which is exactly why it failed with the same token. Upgrading the SDK in the browser-use venv closes the gap.

The change (2 files, testing only)

File	Edit
<code>installed/browser_use.py</code>	Accept <code>AWS_BEARER_TOKEN_BEDROCK</code> in the auth gate; forward it + <code>AWS_REGION</code> / <code>AWS_DEFAULT_REGION</code> (and optional <code>BROWSER_USE_BEDROCK_MODEL</code> / <code>BROWSER_USE_USE_BEDROCK</code>) into the container; install <code>boto3</code> and upgrade <code>anthropic>=0.115</code> in the browser-use venv.
<code>installed/browser_use_runner.py</code>	New <code>_use_bedrock()</code> + <code>_bedrock_model_id()</code> helpers and a <code>ChatAnthropicBedrock</code> branch in <code>_create_llm()</code> . Bedrock is chosen when a bearer token is present and no direct key is set (direct key still wins). Bare aliases (e.g. <code>claude-haiku-4-5</code>) map to the us inference-profile id.

```
.env.local AWS_BEARER_TOKEN_BEDROCK, AWS_REGION (harbor --env-file, gitignored)
└─▶ browser_use.py forwards Bedrock env into the container + upgrades anthropic>=0.115
    └─▶ browser_use_runner.py _use_bedrock() → ChatAnthropicBedrock(model=us.anthropic..., aws_region=...)
        └─▶ anthropic 0.115 reads AWS_BEARER_TOKEN_BEDROCK → Bearer auth → Bedrock round-trip
```

Live trial evidence

2

live Playground trials run through Harbor → Docker



provider=anthropic_bedrock
model authenticated & replied

0

Bedrock auth errors after the SDK upgrade (was 6/6)

Haiku

claude-haiku-4-5
on us inference profile

Trials `pg-game-starclash-62b197c8` (SigV4 failure) and `pg-game-starclash-626b8d45` (Bedrock success) ran the full pipeline: Playground API → Harbor batch job → Docker image build → container → browser-use 0.13.6 → Chromium → verifier. Persona `0164` (Mei Lin Tan) was sampled; player-id wiring fell back to `auto` since the sampled persona isn't in the 4-seat crew manifest.

What is *not* fixed by this (out of scope / known)

Item	Status
Bedrock authentication for browser-use	FIXED
Haiku structured-output adherence (<code>AgentOutput</code> "action field required")	MODEL QUIRK — Haiku returned <code>thinking</code> without <code>action</code> on steps 1–5; a stronger model (Sonnet) or a fallback LLM avoids it.
SPA served to the generic web agent ("page never loaded, 0 screenshots")	ORCHESTRATION GAP — the generic <code>persona-browser-use</code> agent expects a served web app to navigate to; game-starclash renders its HUD via its own <code>--brain vision</code> path, not a hosted SPA URL.
Real multi-agent game (N sampled personas in one match)	NOT YET — unchanged; still needs a Playground orchestration change. Other seats remain <code>BayesianBotBrain</code> bots.

⚠ **Testing-only, do not ship as-is.** Upgrading `anthropic` past browser-use's pin is fine for a local Bedrock test but should be validated against browser-use's own API before landing. These edits live in the shared agent adapter (`environment/agents/matraix/agents/installed/`), **not** in the game-starclash task, and are intentionally kept out of PR #266. Credentials were placed in a gitignored `.env.local` and are never committed.

Bottom line

The question was "*why can't browser-use run on AWS Bedrock?*" Answer: it can — this repo's adapter just never wired up a Bedrock client, and browser-use's pinned `anthropic` SDK was too old to use a Bedrock bearer token. A Bedrock branch in the runner + forwarding the bearer token + upgrading the SDK in the venv makes the Dockerized agent authenticate and run on Bedrock. The genuine game-starclash *gameplay* is still proven separately by the direct `--brain vision` Bedrock run; the two remaining live-trial failures are a Haiku output-format quirk and the pre-existing served-SPA orchestration gap, neither of which is an authentication problem.

browser-use Bedrock testing · agent adapter change (not in PR #266) · trials `pg-game-starclash-62b197c8` & `-626b8d45` · seed 42 · model `claude-haiku-4-5` on us Bedrock inference profile.